

Motion Correction of Interactive CG Avatars Using Machine Learning

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ABSTRACT

Motion capture allows users to control their CG avatar via their own movements. However, the composed avatar motion fails to deliver the actual input movements if the user's motion information is not accurately captured due to measurement errors. In this paper, we propose a method that complements a user's motion according to the motion of another person for a two-party motion with interaction. This method is expected to compose avatar motions that look natural to the other person while emphasizing the actual motions of the user.

Keywords: Interaction, Metaverse, CG Avatar.

Index Terms: Computing methodologies—Computer graphics—Animation—Motion capture; Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Computing methodologies—Machine learning—Learning paradigms—Supervised learning—Supervised learning by regression

1 INTRODUCTION

CG avatars, which serve as a substitute image for the user, have been used in various fields such as virtual reality communication contents including interaction between avatars in a metaverse. Users use motion capture to control their avatars, which respond to their own movements. However, this procedure is hampered, resulting in a failed avatar motion, when inaccurate information is captured because of factors such as the influence of the measurement environment and measurement errors.

A method has been proposed to solve this problem and compose avatar motions with reduced postural failure by complementing the joint poses that are estimated to fail with similar postures captured in advance from the input posture information [1]. In this method, the similar postures used for complementation are calculated from the posture information of the user's joint poses that have not failed. Therefore, if there are only a few joint poses that have not failed, the information for calculating the similar postures will be reduced. As a result, the appropriate similarity posture may not be calculated successfully.

On the other hand, in a two-party motion with interaction, the user's motion has a correspondence with another person's motion. Therefore, by using the pose information of the other person to calculate the similar pose that complements the user's pose, it is expected that an appropriate similar pose can be calculated.

Our method corrects the avatar's motion using the aforementioned interaction. In this method, a motion inference network is constructed by learning the correspondence between the user's and the other person's motions, for an interactive two-party

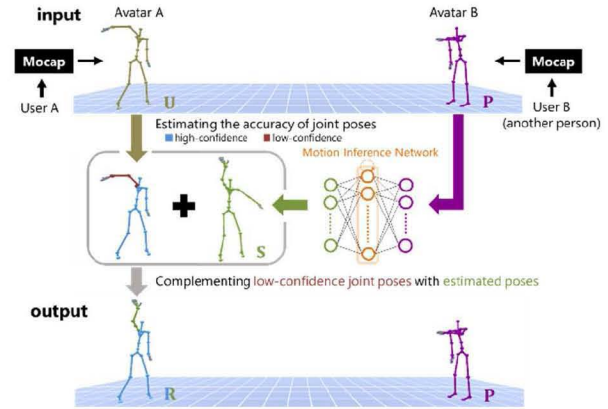


Figure 1: Our Approach

motion. This constructed network is then used to estimate the motion corresponding to the motion of another person. The estimated motion complements the motion of the failed joint poses in the user's motion information. Thus, the avatar's motion that looks natural to the other person is composed while emphasizing the user's actual motions (Fig. 1).

2 OUR APPROACH

2.1 Pre-processing

A motion inference network was constructed using the user's and another person's motions using an interactive two-party motion as training data for machine learning.

In our approach, the motion inference network was constructed using a long short-term memory model [2]. The number of units in the input, middle, and output layers was 75 (rotations of 25 joints and three axes of the other person's whole body), 1000, and 75 (rotations of 25 joints and three axes of the corresponding user's whole body), respectively. The loss function was the mean squared error, and the time series interval was 30 frames of motion data captured at 30fps.

2.2 Main Process in Real Time

First, the user's motion $\mathbf{U} = \{\mathbf{u}^j | j \in M^{input}\}$ and the other person's motion $\mathbf{P} = \{\mathbf{p}^j | j \in M^{input}\}$ were captured by markerless motion capture with Azure Kinect. Next, the confidence of each joint pose of the captured user's motion $\mathbf{U}$ was estimated based on the continuity of posture transitions and the joint rotation limits [1]. The joint poses that were estimated to be correctly captured were defined as high-confidence joint poses M^{conf} , and those estimated to be incorrectly captured were defined as low-confidence joint poses M^{conf} .

Then, the motion information $\mathbf{P}$ of the other person was inputted into the motion inference network, and the motion $\mathbf{S} = \{\mathbf{s}^j | j \in M^{output}\}$ corresponding to the other person's motion was estimated. These estimated motions complemented the user's low-confidence joint poses, and the high-confidence joint poses reflected the input motions directly in the avatar. In this way, we composed an avatar motion $\mathbf{R} = \{\mathbf{r}^j | j \in M^{output}\}$ that looked

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natural to the other person while emphasizing the user's actual motion (Equation 1).

$$\mathbf{r}^j = \begin{cases} \mathbf{u}^j & (j \in M^{conf}) \\ \mathbf{s}^j & (j \in \overline{M^{conf}}) \end{cases} \quad (1)$$

In addition, since our method complemented the user's low-confidence joint poses with motions estimated from the other person's motions, arbitrary joint poses were considered to be low-confidence and were complemented with the estimated motions.

3 RESULTS

We conducted an experiment using a ball-catching motion in order to confirm the correspondence between the motions of the two parties. The training data used for training the machine learning algorithm were 1,235 frames of the ball-catch motion between the user and the other person, each acquired at 30 fps.

First, we verified the complementation of the failed joint poses. The captured data of the user's ball-catch motion $\mathbf{L}$ and the other person's ball-catch motion $\mathbf{P}$, which were not included in the training data, were considered to be the correct answer. Each motion consisted of 438 frames of posture captured at 30 fps. The input motions were the user's motion $\mathbf{U}$ and the other person's motion $\mathbf{P}$. The user's motion $\mathbf{U}$ was the motion corresponding to when the right arm failed, and the motion other than that of the right arm was the same as the correct answer motion $\mathbf{L}$. Additionally, we compared the output motion $\mathbf{R}$ with the correct answer motion, $\mathbf{L}$. The output results are shown in Figure 2. Although the right arm failed in the input motion, the failure of the output motion was verifiably reduced by complementing the right arm's motion with the motion estimated from the other person's motion. A comparison between the correct answer motion and the output motion is shown in Figure 3. The output motion was more similar to the correct answer motion when using our method when compared to using the conventional method [1].

Finally, the user-specified joint poses that might be presented to the other person and other joint poses were used as masked joint poses to compose the avatar motion. The user's input motion $\mathbf{U}$ was the ball-catch motion in a sitting posture. The output result of setting the joint poses other than the head and left arm as masked joint poses are shown in Figure 4, confirming that the masked posture is complemented, and the ball-catch motion of the avatar in the standing postures is composed of the input motion of the partial body in the sitting posture.

4 CONCLUSIONS

We propose an interaction-aware avatar motion correction method for interactive two-party motions. The experimental results confirmed the composed avatar motion to have reduced postural failures. In addition, the composition of the avatar motion of the user's whole-body posture corresponding to the other person's motion was demonstrated using the user's partial posture input. Therefore, our method can be used for composing avatar motions with guaranteed anonymity.

A limitation of our method is that it assumes that the motion information of the other avatar is appropriate. For example, if the other avatar's motion information were to fail, it may not be possible to estimate an appropriate similar posture to use for complementation. Therefore, it is a future challenge to study a method for calculating similar postures that adaptively use the user's motion information and context information in a virtual reality environment in addition to the other person's motion information.

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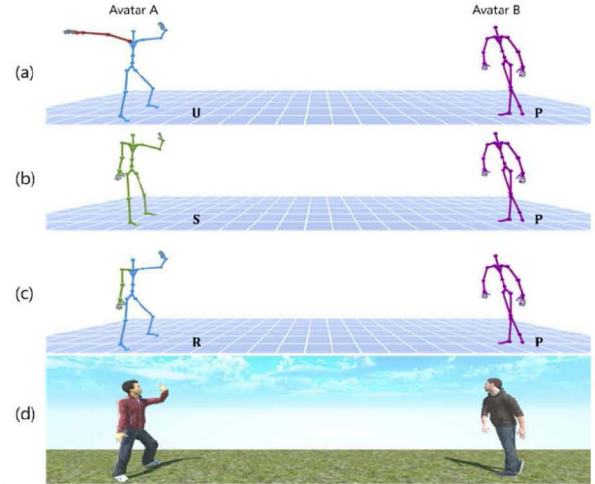


Figure 2: Result of complementation of the failed joint poses. (a) Estimated the confidence of the joint poses. Blue indicates the high-confidence joint poses. Red indicates the low-confidence joint poses. (b) Estimated the motion corresponding to the motion of another person. (c) Complemented low-confidence joint poses with estimated poses. (d) Applied to character model.

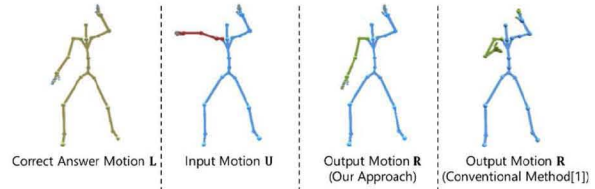


Figure 3: Comparison of output motion and correct answer motion from the other person's point of view.

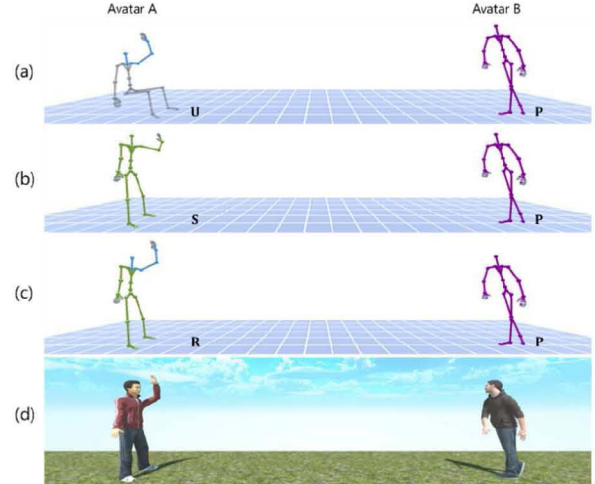


Figure 4: Result of complementation of the masked joint poses. (a) Estimated the confidence of the joint poses. Gray indicates the masked joint poses. (b) Estimated the motion corresponding to the motion of another person. (c) Complemented masked joint poses with estimated poses. (d) Applied to character model.

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